Discrete Mathematics – Final Exam

2022 – Moed B

Instructions for students:

1. In the exam there are **12** short questions and **2** open questions. You must answer **all** of them (No choice).
2. Write your solutions in the attached papers only.
3. When asked to write explicitly a "set/function/relation", you must specify all the elements of the set/function/relation.
4. Use of any external aiding material is **forbidden**. For your convenience, a formula sheet is attached.
5. The duration of the exam is **three hours.**

Good luck!

Part A – Short Questions

- Write your answers clearly in the marked boxes.
- No need to explain how you got to the final answer, **unless it is explicitly asked**.

In counting questions, you can use any of the convenient notations which we used in class, **except** for the symbol $\sum$ and three dots, '…'.

Question 1 (6 points)

For each part, choose which holds: $\in$, $\subseteq$, both, or neither.

i. $\emptyset$ ___ $P(\emptyset)$

ii. $P(\emptyset)$ ___ $P(\mathbb{N})$

iii. $P(\emptyset)$ ___ $P(\mathbb{N} \cup \{\emptyset\})$

Both $\subseteq$ Both

Question 2 (6 Points)

A tautology is a statement whose truth table is full of T's.

Write down the number of the correct claim:

1. if α, β are <u>not</u> tautologies, then $\alpha \to \beta$ is a tautology.
2. if $\alpha \leftrightarrow \beta$ is a tautology, then $\alpha \vee \beta$ is a tautology.
3. If $\alpha \wedge \beta$ is a tautology, then α, β are tautologies.
4. If $\alpha \vee \beta$ is a tautology, then α, β are tautologies.

3

Question 3 (6 points).

Find sets A, B, C such that $A \subseteq B \Delta C$, $A \subseteq C$, and $B \subseteq C$, or briefly explain why no such sets exist.

$A = B = C = \emptyset$

Question 4 (6 points).

What is the coefficient of x^7 in the expansion of the expression $(2+3x+x^2)^6$?

$$\binom{6}{5}\cdot 3+\binom{6}{3,2,1}\cdot 2\cdot 3^3+\binom{6}{3,2,1}\cdot 2^2\cdot 3$$

Question 5 (6 points)

How many ways are there to distribute 50 gold coins to 6 students, such that Arik, Bentz, and Shimon have 21 between them, and every one of the 6 students receives an <u>odd</u> number of coins?

$$\binom{11}{2}\binom{15}{2}$$

Question 6 (6 points)

Suppose $f:\mathbb{N}\to\mathbb{N}$ has a left inverse. Without knowing f, which one of the following $g:\mathbb{N}\to\mathbb{N}$ <u>could</u> be a left inverse of f?

i. $g_1(x)=2x$

ii. $g_2(x)=\begin{cases} x/2 & x \text{ is even} \\ 0 & x \text{ is odd}\end{cases}$

iii. $g_3(x)=x^2-1$

g_2

Question 7 (6 points)

How many "hands" of 5 cards are there from a standard deck of cards, such that the hand has at least one Ace, and no King?

Recall: Standard card deck is 52 cards, with 4 (different) Aces and 4 (different) Kings.

$$\binom{48}{5} - \binom{44}{5}$$

Question 8 (6 Points)

Let $A = \{1,2,3,4,5\}$. Consider relation $R = \{(a,b) \in A \times A : a + b \text{ is even}\}$ on A.

Which of the following properties hold for R? (List all that hold):

Reflexive, Anti-reflexive, Symmetric, Anti-symmetric, Asymmetric, Transitive

Reflexive

Symmetric

Transitive

Question 9 (6 Points).

Let $A = \{1,2,3,4,5\}$. Let S be the following equivalence relation over $A \times A$:

$$(x,y)S(z,w) \iff |y - x| = |w - z|$$

Describe the quotient set $(A \times A)/S$.

$$(A \times A)/S = \{[(1,1)]_S, [(1,2)]_S, [(1,3)]_S, [(1,4)]_S, [(1,5)]_S\}$$

Question 10 (6 points)

How many ways are there of seating 10 people in a circle, such that Bob does not sit next to Alice?

$$9! - 2! \cdot 8! = 7 \cdot 8!$$

Question 11 (6 points)

Let T be the following strong order relation over $\mathbb{R}$:

$$xTy \iff x - y \geq 1$$

Give 2 examples of infinite anti-chains.

$$\{x \in \mathbb{R} | 0 < x < 1\}$$

$$\left\{\frac{1}{n+2} | n \in \mathbb{N}\right\}$$

$$\left\{\frac{1}{(n+1)^2} | n \in \mathbb{N}\backslash\{0\}\right\}$$

$$\left\{\frac{1}{2n} | n \in \mathbb{N}\backslash\{0\}\right\}$$

Question 12 (6 points)

Medical residents A, B, and C and hospitals 1, 2, and 3 have each ranked the other side, as below. Each hospital has one position available.

A: 1, 3, 2	1: C, A, B	(i.e, A prefers 1 most, 3 second, etc)
B: 3, 1, 2	2: A, C, B	
C: 1, 2, 3	3: C, B, A	

Does there exist a stable matching such that A is paired with hospital 1?

If yes, give such a stable matching. If not, explain in one sentence why not.

No.

1 is the first choice of C and vice versa, so 1-C will be a match in any stable matching.

Part B – Open Questions

- Write your answers clearly in the marked boxes. You must prove your answers.
- **Write clear and readable answers! Your grade will also depend on the clarity of your proof and not only the correctness of your answer!**

Question 13 (14 points)

Define the following sequence: $a_0 = 0, a_{n+1} = \frac{1}{4(1-a_n)}$.

Prove that $0 \leq a_n \leq \frac{1}{2}$ for every $n \in \mathbb{N}$.

Use induction.

Base case $n = 0$:

We have $0 \leq 0 = a_0 \leq \frac{1}{2}$.

Inductive assumption:

Assume $0 \leq a_n \leq \frac{1}{2}$.

Inductive step:

We have $0 \leq a_n \leq \frac{1}{2}$, hence

$$0 \leq a_n \leq \frac{1}{2}$$

$$-\frac{1}{2} \leq -a_n \leq 0$$

$$\frac{1}{2} \leq 1 - a_n \leq 1$$

$$2 \leq 4(1 - a_n) \leq 4$$

$$0 \leq \frac{1}{4} \leq \frac{1}{4(1 - a_n)} \leq \frac{1}{2}$$

$$0 \leq a_{n+1} \leq \frac{1}{2}$$

As desired.

Question 14 (14 points)

Let $f, g: \mathbb{N} \to \mathbb{N}$ be total functions. Assume $f(n) = 2g(n) - 1$ for every $n \in \mathbb{N}$.

a. Prove f is not onto.

b. Prove that if g is onto, then $Im\ f = \mathbb{N}_{odd} \cup \{-1\}$.

Remark: Recall that for $f: A \to B$, we define $Im\ f = \{b \in B | \exists a \in A\ (f(a) = b)\}$.

a. Take $0 \in \mathbb{N}$. ABC that there exists $n \in \mathbb{N}$ such that $f(n) = 0$. Hence

$$f(n) = 0$$
$$2g(n) - 1 = 0$$
$$g(n) = \frac{1}{2} \notin \mathbb{N} = Codomain(g)$$

a contradiction.

b. Let $m \in Im\ f$. So there exists $n \in \mathbb{N}$ such that $f(n) = m$. Hence $2g(n) - 1 = m$.

But $g(n) \in \mathbb{N}$, so $m = 2g(n) - 1 \in \mathbb{N}_{odd} \cup \{-1\}$. Hence $Im\ f \subseteq \mathbb{N}_{odd} \cup \{-1\}$.

Let $m \in \mathbb{N}_{odd} \cup \{-1\}$. We have $m + 1 \in \mathbb{N}_{even}$, hence $\frac{m+1}{2} \in \mathbb{N}$. g is onto, so there exists $n \in \mathbb{N}$ such that $g(n) = \frac{m+1}{2}$. Hence $f(n) = 2g(n) - 1 = 2\frac{m+1}{2} - 1 = m$.

Hence $m \in Im\ f$. Hence $\mathbb{N}_{odd} \cup \{-1\} \subseteq Im\ f$.

We conclude that $Im\ f = \mathbb{N}_{odd} \cup \{-1\}$.

An extra page for the open questions:

Discrete Mathematics – Formula sheet for the exam

Set Theory:

$$A\Delta B = (A\backslash B) \cup (B\backslash A)$$
$$A \times B = \{(a,b) |\ a \in A\ and\ b \in B\}$$
$$P(A) = \{S | S \subseteq A\}$$

$$|A \times B| = |A| \cdot |B|$$
$$|P(A)| = 2^{|A|}$$

De-Morgan:

$$\overline{B \cup C} = \bar{B} \cap \bar{C} \qquad \mathrm{A}\backslash(B \cup C) = (A\backslash B) \cap (A\backslash C)$$
$$\overline{B \cap C} = \bar{B} \cup \bar{C} \qquad \mathrm{A}\backslash(B \cap C) = (A\backslash B) \cup (A\backslash C)$$

Distributive:

$$\mathrm{A} \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$\mathrm{A} \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Functions/Relations:

$$i_{\mathrm{A}} = \{(a,a) | a \in A\}$$

Logic:

For all formulas α, β, γ :

De-Morgan:

$$\neg(\alpha \wedge \beta) \equiv \neg\alpha \vee \neg\beta$$
$$\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$$

Distributive:

$$\alpha \wedge (\beta \vee \gamma) \equiv (\alpha \wedge \beta) \vee (\alpha \wedge \gamma)$$
$$\alpha \vee (\beta \wedge \gamma) \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$$

Negation rules:

$$\neg\forall \mathrm{x}(\alpha) \equiv \exists x(\neg\alpha)$$
$$\neg\exists \mathrm{x}(\alpha) \equiv \forall x(\neg\alpha)$$

Combinatorics:

Choosing k items out of n different items:

	With repetition	Without repetition
With order	n^k	$\frac{n!}{(n-k)!}$
Without order	$\binom{k+n-1}{n-1}$	$\binom{n}{k} = \frac{n!}{(n-k)! \cdot k!}$

Binomial Formula

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} \cdot x^k y^{n-k}$$

Multinomial Coefficient:

$$\binom{n}{i_1, i_2, i_3, \dots, i_k} = \frac{n!}{i_1! \cdot i_2! \cdot i_3! \cdot \dots \cdot i_k!} \quad if \;\; i_1 + i_2 + i_3 + \cdots + i_k = n$$

Multinomial Formula:

$$(x_1 + x_2 + x_3 + \cdots + x_k)^n = \sum_{i_1+i_2+i_3+\cdots+i_k=n} \binom{n}{i_1, i_2, i_3, \dots, i_k} \cdot x_1^{i_1} \cdot x_2^{i_2} \cdot x_3^{i_3} \cdot \dots \cdot x_k^{i_k}$$

Pascal's Triangle:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

$$\binom{n}{k} = \binom{n}{n-k}$$

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$$

Inclusion-exclusion principle:

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{I \subseteq \{1,2,\dots,n\},\;\; I \neq \emptyset} (-1)^{|I|-1} \cdot \left| \bigcap_{\mathrm{i} \in \mathrm{I}} \mathrm{A_i} \right|$$

Generalized Pigeonhole Principle:

If we put m pigeons in n pigeonholes, then there is at least one pigeonhole in which there will be at least $\left\lceil \frac{m}{n} \right\rceil$ pigeons.